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Social support, age, and emotion regulation predict symptom severity of adults with major depressive disorder

Alyssa M. Nelson[1], Joseph M. Diehl[1], Kevin S. LaBar[1], and Moria J. Smoski[1,2]

[1]Duke University, [2]Duke University Medical Center

Abstract – The aim of this study was to examine the relationships among social support, age, emotion regulation, and depressive symptoms in adults with Major Depressive Disorder (MDD). A better understanding of remediable factors that are related to MDD outcomes could inform effective treatments and preventative strategies. A total of 42 participants aged 35-68 responded to survey items assessing social support, distraction use, reappraisal use, and depressive symptom severity. Depression severity was more strongly associated with lower social support for older participants. Lower emotion regulation strategy use was associated with greater depressive symptoms, a relationship that was accounted for by lower social support. Social support is a key contributor to the positive impact of emotion regulation ability on mitigating depression symptoms in mid-life to older adults with MDD. Emotion regulation-based interventions may benefit from a focus on enhancing social support and positive social interactions. Future research should utilize experimental manipulations and longitudinal data analysis to establish causality in the observed relationships.

Keywords: adulthood, emotion regulation, Major Depressive Disorder, social support

Major Depressive Disorder (MDD) is a pervasive mental health condition with significant adverse consequences across the adult lifespan. About 20% of American adults experience a depressive episode in their lives, rendering MDD one of the most common mental health conditions in the United States (Hasin et al., 2018). In addition, MDD has significant detrimental consequences for those affected as well as society at large (Charlson et al., 2013; Lawrence et al., 2022; Russo et al., 2015; Pérez-Sola et al., 2021). The negative ramifications of depression generalize beyond the realms of emotional well-being and daily functioning. A robust literature has demonstrated that depression has a substantial negative impact on somatic health, increasing risk of morbidity and mortality (Penninx et al., 2013; Chiu et al., 2018).

Effective interventions for depression are vital due to its prevalence and devastating consequences. Although prior research supports the effectiveness of various treatments for MDD (Cuijpers et al., 2023), available therapies have limitations with respect to initial treatment response (Lemmens et al., 2011) and relapse (Paykel, 2008). A better understanding of factors that are related to depression outcomes could inform effective treatments and preventative strategies for MDD. Positive relationships and connections with others, as well as the ability to intentionally regulate one's emotions, are known to mitigate depression severity (Leskela et al., 2006; Campbell-Sills et al., 2006; Daros et al., 2023). However, the relationship between social support and emotion regulation, especially in older adults, is less clear.

A robust literature confirms that social support is linked to positive clinical outcomes for those with MDD. Subjective perceptions of strong social support, as well as frequent social interactions, are associated with a lower risk of developing a recurrent episode of MDD and reductions in depressive symptoms over time (Sherbourne et al., 1995; Leskela et al., 2006). Research has also found social support to be associated with favorable treatment outcomes for those undergoing therapy for MDD (Solomonov et al., 2019). Unfortunately, the size of social networks tends to diminish as individuals progress through adulthood (Schnittker, 2007). This reduction occurs as social support may be more salient, with one study demonstrating that the protective effects of social support for onset of MDD and the clinical course of the disease were most pronounced for older patients (Sherbourne et

al., 1995). It is unclear if other age-related changes in cognition and emotion may impact the relationships among social support, age, and depression.

Emotion regulation is another factor that has important implications for MDD. Impaired emotion regulation is a characteristic feature of those who have MDD (Joormann & Stanton, 2016). Specifically, individuals with MDD seem to have greater difficulty regulating negative affect after adverse life experiences, and this has been found to be implicated in the onset of many individuals' first depressive episode (Hammen, 2005). Additionally, those with MDD report less frequent use of effective emotion regulation strategies such as reappraisal (thinking differently about a negative stimulus to alter emotional response) (Joormann & Stanton, 2016). Clinical populations who are struggling with depression also derive less benefit from some emotion regulation strategies (Smoski et al., 2014) and are often unable to return to their emotional baseline as quickly as non-clinical populations (Joormann, 2010). This may be due in part to the fact that those with MDD have increased ruminative thinking, meaning their thoughts tend to recur, or "recycle" themselves (Flett et al., 2002; Spasojevic & Alloy, 2001). This pattern of thinking is particularly detrimental in the context of depressed moods, as rumination enhances negative thoughts, interferes with instrumental behavior that could improve mood, and increases the likelihood of social rejection during depressive episodes (Nolen-Hoeksema et al., 2008).

To date, few studies have explored the relationship between social support and emotion regulation in MDD. One theoretical link between the constructs of social support and emotion regulation is Social Baseline Theory, which posits that violations of an expectancy of social support tax cognitive and affective resources (Coan & Sbarra, 2015), such that fewer resources are available for emotion regulation in the context of low social support. Conversely, strong social support seems to facilitate attempts to regulate emotion. Morawetz et al. (2021) demonstrated that reappraisal strategies in response to disturbing images are enhanced with the support of a friend compared to the support of a stranger or when alone. Thus, the inverse relationship between social support and depression may be partially explained by emotion regulation.

The aim of the present study was to more comprehensively examine the relationships among social support, age, emotion regulation, and depressive symptoms in middle aged to older adults with MDD. We hypothesized that the negative association between social support and depressive symptoms would get stronger as age increased in adulthood. In addition, we hypothesized that effective emotion regulation strategy use would at least partially account for the relationship between social support and depression.

Methods

This study is a secondary analysis of an observational study examining the neurobehavioral mechanisms of emotion regulation across the adult lifespan [NCT03207503]. The parent study protocol was approved by the Duke Health IRB. Some participants included in the present study's analysis did not complete the entire protocol for the parent study. Their data was included if they met screening, clinical, and neuropsychosocial requirements.

Participants

Participants included 42 adults diagnosed with MDD. The mean age in the sample was 49.67 (*SD* = 10.71, range = 35-68). The majority of participants identified as female (86%) and white (79%). Black or African American participants made up 12% of the sample, with 5% who identified as Hispanic/Latinx. Inclusion criteria included the following: not currently pregnant, ambulatory with no known uncorrected sensory deficits, an estimated verbal IQ of at least 88, and a Beck Depression Inventory-2 (BDI-2; Beck et al., 1996) score of at least 14 with current MDD or Persistent Depressive Disorder (PDD) with either persistent major depressive episode or intermittent major depressive episodes. Exclusionary criteria included current substance use disorder; current or history of psychosis, mania, or eating disorder; disorder with impact on brain characteristics; contraindications to MRI scanning; unstable medicine regimen of antidepressants, antipsychotics, or mood stabilizers; and indication of mild cognitive impairment or dementia. Participants were recruited through online advertisements, letters from their primary care physicians, and research recruitment databases, including the Duke Brain Imaging and Analysis Center control subject registry, ResearchMatch, and the Duke Cognitive-Behavioral Research Treatment Program participant registry.

Procedure

The current study is a secondary data analysis in which existing data from a parent study was analyzed to address a novel research question. The parent study of the current project was an observational study that incorporated interview, behavioral, and neuroimaging measures of depression as well as survey measures of social support and emotion regulation. Data for the present study was drawn from the aforementioned survey responses. Participants completed surveys online using

REDCap (Research Electronic Data Capture), an online application that collects and stores data for clinical research projects (Patridge & Bardyn, 2018). They also completed surveys in-person at Duke University. Due to COVID-19 restrictions, some participants also completed components of the study remotely over the phone or online over Zoom or WebEx.

Participants completed a pre-screen over the phone or through REDCap (based on participant preference) to determine potential study eligibility. Those who were eligible were directed to an online consent form hosted by REDCap. Once the participants had given informed consent, they were scheduled for up to two assessment visits to further determine study eligibility. Assessments included clinical interview and a brief neuropsychosocial screener. Eligible participants were then asked to respond to survey questions administered through REDCap that assessed demographics, social support, and emotion regulation. Two more study visits, which involved data that will not be analyzed in this study, were scheduled after this.

Measures

MDD diagnosis was confirmed via the *Structured Clinical Interview for DSM-5–Research Version* (SCID-5-RV; First et al., 2015), a structured interview to assess lifetime psychiatric disorders and to document duration, onset, and number of depressive episodes. Depression severity was indexed by the *Beck Depression Inventory-II* (BDI-II; Beck et al., 1996), a 21 item, self-report instrument that measures severity of depressive symptoms. Cronbach's α for this measure was .97, suggesting acceptable internal reliability. A total composite score of the *Duke Social Support Index* (DSSI; Koenig et al., 1993) indexed social support. The DSSI is a 23 item, self-report instrument that measures social networks, social interactions, subjective support, and instrumental support. Cronbach's α for this measure was .89, suggesting acceptable internal reliability. Emotion regulation was measured as a summed standard score of distraction use (the 11 distraction items of the *Response Styles Questionnaire* (RSQ; Nolen-Hoeksema & Morrow, 1991)) and reappraisal use (the 6 reappraisal items of the *Emotion Regulation Questionnaire* (ERQ; Gross & John, 2003)). Cronbach's α for this measure was .89, suggesting acceptable internal reliability. Finally, participants provided information on gender, age, race, ethnicity, education, city of birth, and marital status.

Analyses

A summary emotion regulation variable was created by summing the Z-scores of the RSQ-distraction and ERQ-reappraisal scores. Bivariate correlations were conducted to establish associations between social support scores and depression scores and between social support scores and age. Moderation analysis was conducted using PROCESS in SPSS and reverified using the ggplot2 package in R to test whether the association between depression scores and social support became stronger as age increased. Next, bivariate correlations were conducted to establish associations between emotion regulation scores and social support scores and between emotion regulation scores and depression scores. Partial correlations were then conducted between depression scores and social support scores while accounting for summary emotion regulation scores. Finally, partial correlations were conducted between depression scores and emotion regulation scores while accounting for social support scores.

Results

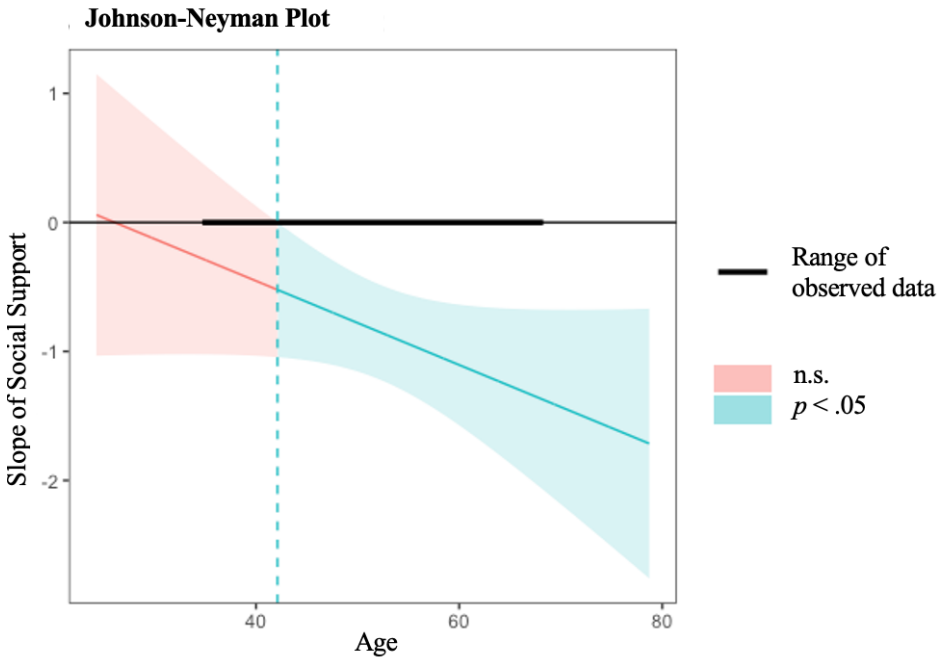
Associations between social support and depression ($r = -.57$, $p < .001$) and between social support and age ($r = -.35$, $p = .02$) were statistically significant. A multiple regression analysis with social support, age, and their interaction term as predictors of depression severity was computed. The overall model was a good fit, $F(3, 42) = 8.46$, $p < .001$, $R^2 = .40$. Higher social support was a significant predictor of lower depression ($b = 0.76$, $p < .01$) but age was not ($b = 0.16$, $p = 0.15$). The interaction term trended toward significance ($b = -0.33$, $p = .08$), and thus, the moderation analysis was probed using the Johnson-Neyman procedure. Beginning at age 42, the effect of age on the association between depression and social support was significant ($b = -.52$, $p < .05$). As age increased from this point, the negative association between social support and depression grew stronger until it reached the highest age in the sample ($b = -1.37$, $p < .001$). A Johnson-Neyman plot of these results is presented in Figure 1. A conceptual model of age as a moderator of the association between social support and depression is depicted in Figure 2.

Associations between emotion regulation and social support ($r = .38$, $p = .01$) and between emotion regulation and depression ($r = -.35$, $p = .02$) were statistically significant. Contrary to our hypothesis, when accounting for emotion regulation, the inverse association between social support and depression severity remained significant ($r = -0.51$, $p < .001$). However, when accounting for social support, the inverse association between emotion regulation and depression became nonsignificant ($r = -.173$, $p = .278$).

Discussion

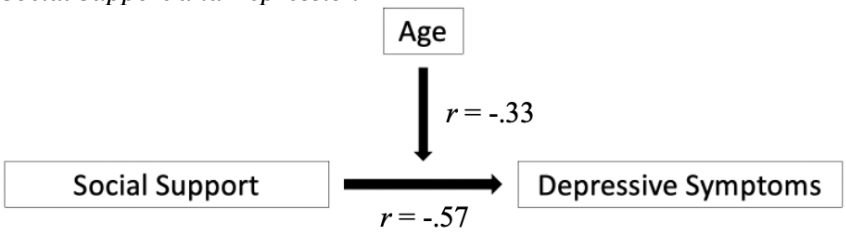
The objective of this study was to examine the relationships among social support, age, emotion regulation, and depressive symptoms in adults with

Figure 1
Johnson-Neyman Plot of Age as a Moderator of the Association Between Social Support and Depression



Note. Beginning at age 42 in our sample, the effect of age on the association between depression and social support was significant. As age increased from this point, the negative association between social support and depression grew stronger until it reached the highest age (68) in the sample.

Figure 2
Conceptual Model of Age as a Moderator of the Association Between Social Support and Depression



MDD. The results suggest that social support and age are negatively associated and that age moderates the negative association between social support and depression. Additionally, findings indicated that social support and use of effective emotion regulation strategies are positively associated. Furthermore, the present study discovered that the negative association between emotion regulation and depression is accounted for by the positive relationship between emotion regulation and social support.

Consistent with hypotheses, greater social support was more strongly related to diminished depressive symptoms for the oldest adults in this sample. These findings are in line with earlier work (Sherbourne et al., 1995), which demonstrated that social support elicited the greatest protective effects for the onset and course of MDD for older adults. It is noteworthy to mention that the present findings also align with prior research indicating an inverse relationship between social support and age. As social networks shrink, the risk of experiencing depression tends to increase (Gianfredi et al., 2021). This underscores the heightened necessity of cultivating resilient social connections and support systems to enhance the mental well-being of older adults.

This study also provides evidence that social support and emotion regulation are positively related, consistent with prior research (Cooke et al., 2019; Morawetz et al., 2021). Specifically, greater social engagement and satisfaction with support received were associated with greater use of distraction and reappraisal strategies. One potential explanation of these findings is that engaging in social interactions and feeling supported by one's social network may provide opportunities to leverage strategies that diminish negative affect. This is consistent with recent research that indicated leveraging social resources is a promising avenue for regulating emotion in MDD (Liu et al., 2024). Alternatively, leveraging strategies to regulate negative emotions may make it easier to form and maintain relationships. A final possibility is that a third variable not measured in the current study could be related to both social support and

emotion regulation, explaining their positive association. For instance, a factor of psychological health may enable individuals to establish and maintain social relationships as well as engage in strategies that diminish their negative emotions.

Concerning depression severity, the present findings indicate a negative association with use of effective emotion regulation strategies. However, contrary to expectations, better emotion regulation did not account for the negative association between social support and depression. Instead, the association between emotion regulation and depression became nonsignificant after accounting for social support. This pattern suggests that leveraging distraction and reappraisal diminishes depressive symptoms by enhancing satisfying social engagement. These results are consistent with Coyne's (1976) empirically supported interpersonal theory (Marcus & Nardone, 1992), which posits that individuals with persistent negative affect are eventually distanced from family and friends, leading to the loss of social support and the onset of depressive episodes.

The present findings have clinical implications, as they provide insight regarding interventions for depression. Consistent with prior research (Sherbourne et al., 1995; Leskela et al., 2006), having a strong social network and feeling supported by this network was associated with fewer depressive symptoms for those with MDD, particularly in older age. This may be related to the idea that social support "buffers" against the impacts of stressors and strengthens the ability to bounce back from stressful life events that might otherwise lead to depression (Bovard, 1959). In addition, the present findings indicate that without strong social support, emotion regulation may not be a protective factor for depressive symptom severity in older depressed adults. Emotion regulation-based interventions may benefit from a focus on enhancing social support and positive social interactions, as demonstrated in the ENGAGE intervention for late-life MDD (Solomonov et al., 2019).

While the present study has notable strengths, such as implementation of rigorous exclusion criteria and control measures during enrollment, it is important to note methodological limitations. The lack of experimental manipulations and longitudinal data analysis prevents establishing causality in the observed relationships. Additionally, the results have limited generalizability to those with marginalized identities, as the sample was predominantly white. Finally, the statistical power of the study was limited due to the small sample size. Future research should address these limitations by increasing the sample size and the diversity of their samples with respect to age, race, ethnicity, and gender. Also, longitudinal and/or experimental methodology would provide greater clarity as to the directionality of the relationships observed among the variables in this study.

Despite the aforementioned limitations, this study contributes to a greater understanding of the intricate relationships among social support, age, emotion regulation, and depression. Most notably, the study emphasizes that social support is a key contributor to the positive impact of emotion regulation ability on mitigating depression symptoms in mid-life to older adults with MDD.

The findings provide future directions for a more nuanced clinical approach to MDD across the adult lifespan. These results underscore the importance of developing therapeutic interventions for MDD that consider age-related factors to address the specific needs of different age groups. For older adults in particular, therapeutic programs for MDD should prioritize bolstering social support through strengthening of existing relationships and fostering new sources of social connection. This approach can enhance emotion regulation abilities and improve mental health outcomes in this demographic.

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Correspondence concerning this article should be addressed to Alyssa M. Nelson, Duke University, 113 Edens Drive, Duke University Box # 99217, Durham NC 27708. Email: Amn50@duke.edu